dragonfly larvae on the behavior and physical development of their tadpole prey.

Since the mid-20th century, many new ideas on ecological processes have emerged. Work by Robert MacArthur and others on competition between species led to the development of optimal foraging theory, which seeks to explain why animals choose to eat some food items and not others. Mutualistic relationships became better understood through the research of biologists such as Daniel Janzen. Robert Paine's work with starfish and mussels also highlighted the concept of keystone species—those that have a disproportionate influence on their ecosystems.

New technology

Technological advances—including sophisticated chemical sampling techniques, satellites with remote sensing equipment, and computers capable of rapidly processing huge quantities of data—have opened up new areas of study.

Ecological stoichiometry, for example, studies the flow of energy and chemical elements throughout food webs and ecosystems, from the molecular level up. Like so many ideas in ecology, its origins can be traced back many years, but only took hold with Robert Sterner and James Elser's 2003 book *Ecological stoichiometry: The biology of elements from molecules to the biosphere*. New techniques such as this will undoubtedly continue to deepen our understanding of processes in ecology.